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EXPERIMENTAL INVESTIGATION OF THE CHEMISTRY OF EXCITED STATES OF RARE GASES

Progress Report for ~~Renewal Proposal~~
Report Period Oct. 15, 1976 - July 1, 1977

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Summary

During this report period, Mr. Harold Brashears (graduate student) has worked full time, Mr. J. E. Velazco completed his Ph.D. dissertation and Dr. Bob Chang (postdoctoral fellow) has worked several months. Mr. Brashears has developed techniques which permit vacuum U.V. photolysis studies of KrF_2 and XeF_2. With these techniques we are able to study the fluorescence and quenching of XeF(B), XeF(D) and KrF(B). The preliminary account of the XeF(B) quenching studies has recently been published and the work with KrF(B) was described in the last quarterly progress report. Mr. Brashears also is setting up the vacuum U.V. monochromator and has done preliminary investigations of the reactions of the resonance ($^3P_1$) states of Kr and Xe with halogens. Dr. Chang's studies utilize the pulsed laser fluorescence technique (see published paper 2 of the following list or earlier quarterly reports for a more complete description of the technique) to generate the excited p states of the rare gases. From these experiments we obtain the radiative lifetimes of the p states, the two-body quenching rate constants and the final product states of the two-body quenching, e.g.,

$$\begin{aligned} Ar(4s, {}^3P_2) + \text{laser pump} &\longrightarrow Ar(4p'\,[3/2]_2) \\ Ar(4p'\,[3/2]_2) + Ar &\longrightarrow Ar(4p'\,[3/2]_1) + Ar \\ &\longrightarrow Ar(4s', {}^3P_1) + h\nu \\ Ar(4p'\,[3/2]_1) &\longrightarrow Ar(4s', {}^3P_1) + h\nu \end{aligned}$$

We have discovered that the product states differ for various Ar(4p') levels; some give intramultiplet cascade, as shown above, while others apparently are quenched to the Ar(4s and 4s') levels. The two-body rate constants are large enough that three-body effects may not be important for many practical laser systems and understanding these

two-body processes could be of some direct importance for optimization of operating conditions for lasers using electrical pumping of rare gases. Dr. Chang also is using the technique to study the Xe(6p') and Xe(7p) states in argon buffer gas. When this work is completed, we expect to have definitive experimental measurements for the radiative lifetimes (and hence absolute oscillator strengths for the spectral lines), two-body rate constants and the associated product state assignments. Such data are of fundamental importance to rare gas chemistry and physics, as well as being of practical value for laser systems.

In addition to the published study of $Xe(^3P_2)$ atoms reacting with various fluorine containing molecules, Dr. Velazco examined an analogous series of chlorine containing molecules for his Ph.D. dissertation. We now are involved in writing manuscripts describing that work, and papers 7 and 8 in the list below will be the end result. Dr. Velazco measured quenching rate constants, branching fractions for $XeCl^*$ formation, the ratios of XeCl(B) to XeCl(C) and the variation of this ratio with pressure and different reagents. Under NSF support, Dr. K. Tamagake of our research group has developed a computer program that simulates the bound-free diatomic emission spectra.[§] Work is in progress to simulate Velazco's $XeCl^*$ spectra from various reagents. When the simulations are completed, we will have a description of the vibrational distributions, as well as some information about the shape of the potential surfaces at high energy. Very little ERDA money has been spent on this project during the Oct. 15,1976 - Oct. 14, 1977 period. However, it was the ERDA funds from the previous year that supported Velazco's experimental measurements which stimulated our efforts to simulate the

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[§]The first application was to the KrF^* emission spectra, J. Chem. Phys., submitted (1977).

bound-free rare gas halide spectra. Therefore, our progress with this work is mentioned here.

In the proposal for this contract year, five projects were identified (and it was noted that not all the work could be done in one year).

(i) Quenching studies and branching ratios from $Ar(^3P_2)$ and $Kr(^3P_2)$ atom reactions with halogen containing molecules and modeling of bound-free spectra.

(ii) Quenching of $Xe(^3P_1)$ and $Kr(^3P_1)$ resonance states by halogens.

(iii) Quenching and lifetime measurements of Xe(6p') and Xe(7p) states.

(iv) Develop methods for studying electronic quenching of XeF(B).

(v) Study metastable atom reactions with oxygen containing molecules in an effort to find a good source of the ionic rare gas oxide excimers.

Because of the success with the photolysis of XeF_2 and KrF_2, no work was started with (v). For the same reason, the work suggested under (i) actually was done with personnel supported by funds other than ERDA. In general the work has progressed rather closely to the plan of the proposal.

Following the cumulative listing of published papers below, the work of Mr. Brashears and Dr. Chang will be summarized briefly. A reprint of the published work with XeF(B) is attached as an appendix. In addition to the work done exclusively in our laboratory, we also wish to mention work done by John Coxon (Dalhousie University) and Michael Clyne (Queen Mary College, University of London) that has evolved from observations made at Kansas State University. Dr. Coxon has characterized new emission spectra from the PCl molecule excited from $Ar(^3P_{0,2})$ reacting with PCl_3. Dr. Clyne has assigned a rate constant to excitation of the H(n=2) from collisions between $Ar(^3P_{0,2})$ and H(n=1).

Cumulative Listing of Published Papers

1. J. E. Velazco, J. H. Kolts and D. W. Setser, "Quenching rate constants for metastable argon, krypton, and xenon atoms by fluorine containing molecules and branching ratios for XeF and KrF formation," J. Chem. Phys. 65, 3468 (1976).

2. D. L. King, J. Rodriguez and D. W. Setser, "Lifetime and two-body rate constants of Ar(4p)** states," Third Summer Colloquim on Electronic Transition Lasers, Aspen, Colorado (Sept. 1976).

3. J. E. Velazco, J. H. Kolts and D. W. Setser, "Quenching rate constants for $Ar(^3P_{0,2})$, $Kr(^3P_2)$ and $Xe(^3P_2)$ by halogen containing molecules and branching ratios for XeF and KrF formation," Twelfth International Photochemistry Conference, Washington, D.C. (July, 1976).

4. J. E. Velazco, J. H. Kolts, D. W. Setser and J. A. Coxon, "Excitation of XeF by the reactions of XeF_2 with $Ar(^3P_{0,2})$, $Xe(^3P_2)$ and $Kr(^3P_2)$," Chem. Phys. Letts. 46, 99 (1977).

5. H. C. Brashears, Jr., D. W. Setser and D. DesMarteau, "Vacuum-ultraviolet Photolysis of XeF_2: XeF(B) Fluorescence and Quenching," Chem. Phys. Letts. 48, 84 (1977).

6. J. E. Velazco "Quenching Reactions of $Xe(^3P_2)$ Reactions," Ph.D. Dissertation Kansas State University (1977).

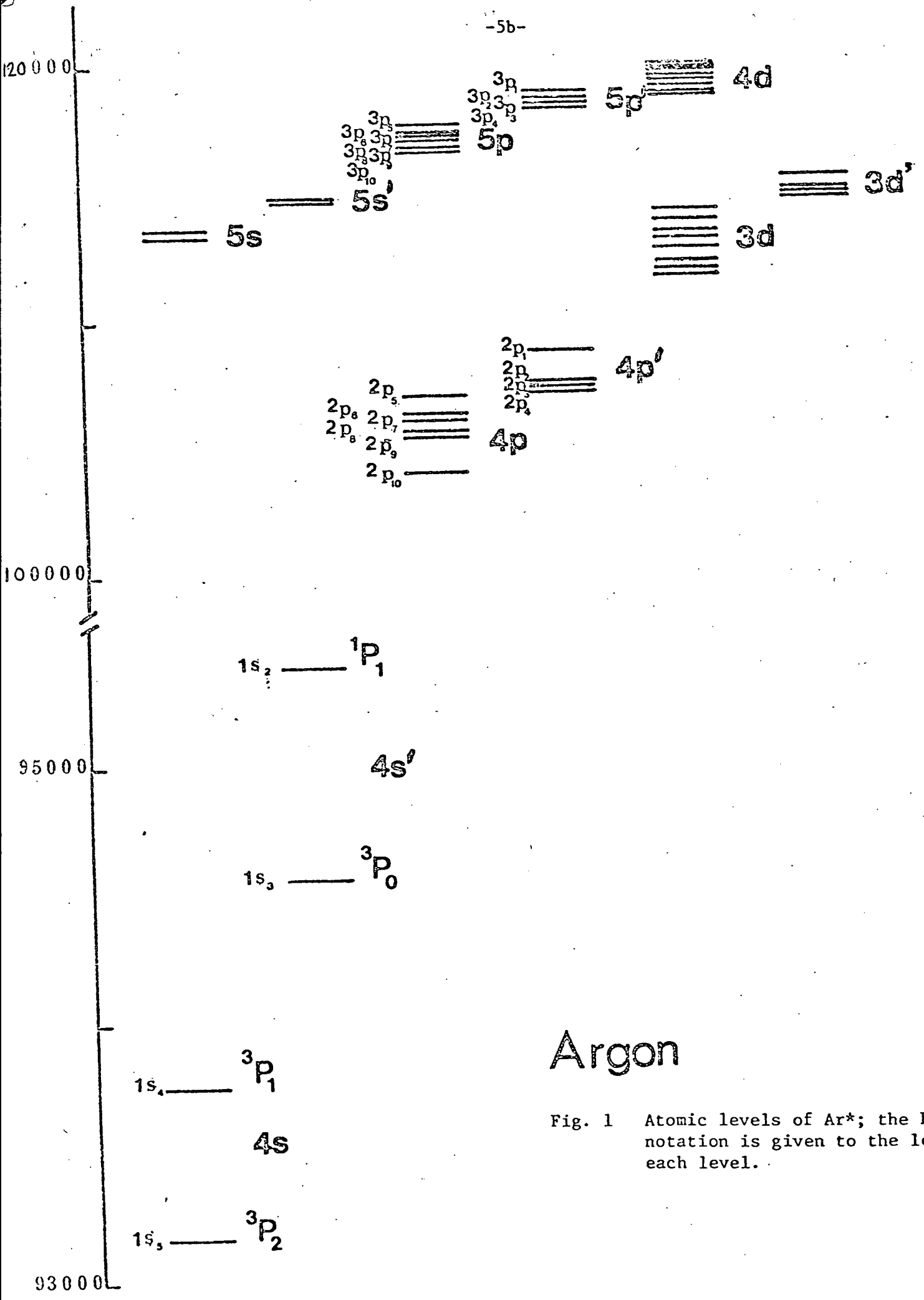
7. J. E. Velazco, J. H. Kolts and D. W. Setser, "Formation of $XeCl^*$ from Reactions of $Xe(^3P_2)$ with Chlorine Containing Molecules," J. Chem. Phys., to be submitted.

8. K. Tamagake, J. H. Kolts and D. W. Setser, "Analysis of the $XeCl^*$ Bound-Free Emission," J. Chem. Phys., to be submitted.

Laser Induced Fluorescence Studies of Ar(4p'), Ar(5p), Xe(6p') and Xe(7p) States.

In our initial studies of the Ar(4p') and Ar(5p) levels, which were reported at the Colloquium on Electronic Transition Lasers at Aspen,[7] we had encountered two problems. For ease of following the discussion of the text, see Figure 1 for identification of the Ar states. The first problem was that our directly measured Ar(4p') radiative lifetimes were longer than the consensus values in the literature. Although the literature values generally were not directly measured, prudence suggested that we check our experimental technique. In doing so, it was discovered that the time constant of our detection gear was modifying the true decay times. Therefore, we replaced the module in the PAR boxcar with one having a 1 nsec gate width. After carefully checking the time constants, we now are confident that the decay times which we measure are true ones. The redetermined radiative lifetimes and two-body rate constants are given in Table 1. The changes are only about 10-15%, but the new values generally are in closer agreement with the consensus literature results. It should be noted that these two-body rate constants for the Ar(4p') levels are sufficiently large that three-body effects will not be important until relatively high pressures.

The second problem was that fluorescence from the Ar(5p) levels could never be observed directly. Instead only the growth and decay of the emission from the Ar(4p) levels could be followed. This is partly a consequence of the unfavorable radiative branching ratios which favor the Ar(3d) and Ar(5s) pathways rather than the Ar(4s) levels. Also, rapid collisional transfer processes may deplete the Ar(5p) level being studied and reduce the fluorescence. For these and perhaps

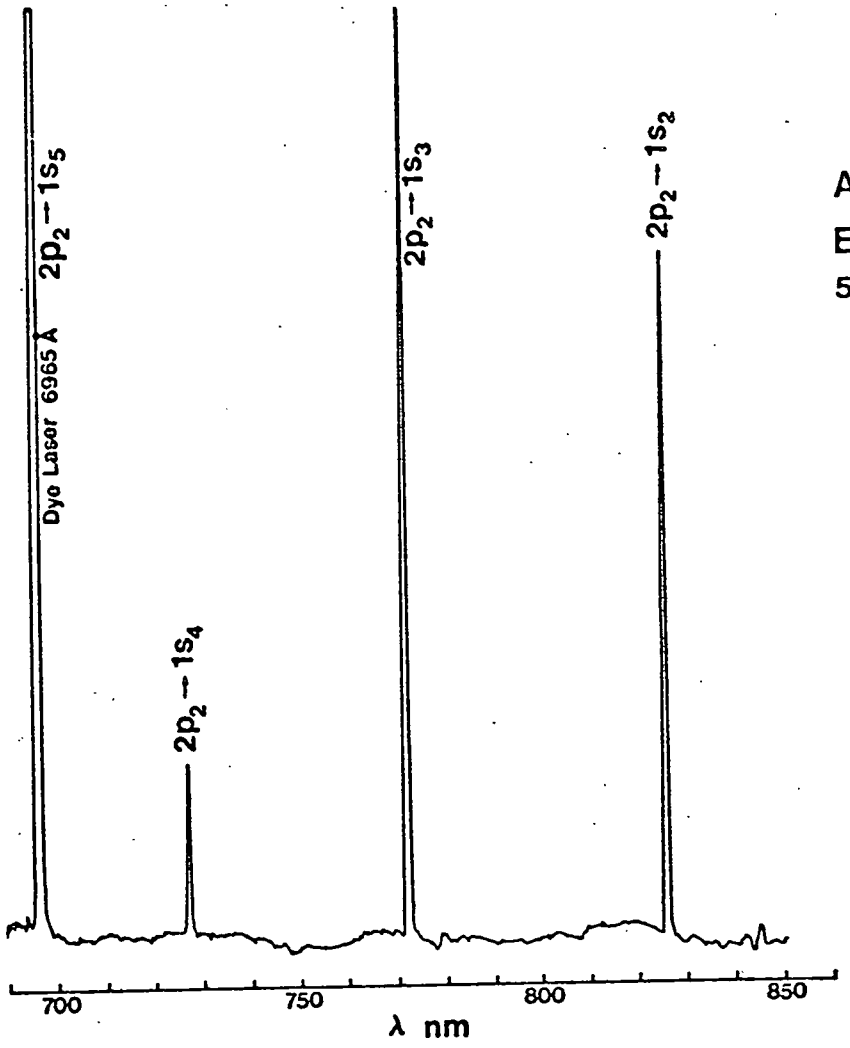


Argon

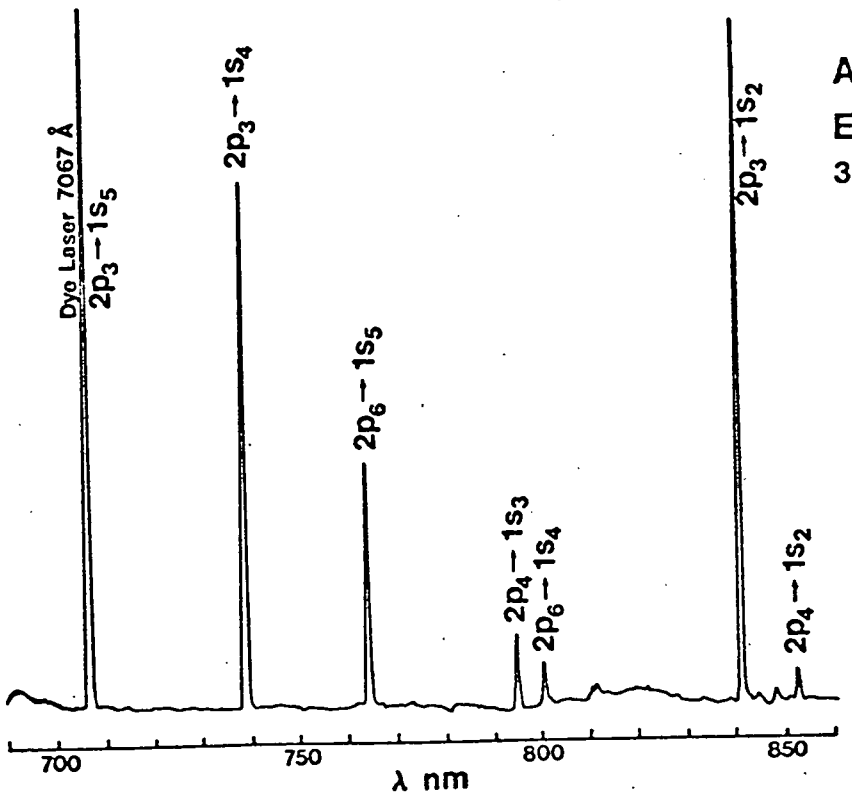
Fig. 1 Atomic levels of Ar*; the Paschen notation is given to the left of each level.

other reasons, we were unable to observe fluorescence from the Ar(5p) levels in the presence of the scattered laser light when using interference filters to observe the fluorescence. During the last six months, improvements have been made which now permit us to observe the fluorescence with a monochromator. With this approach, it has been possible to observe the very weak fluorescence from the Ar(5p) levels. At low pressure the emission from the individual Ar(4p) levels produced by radiative cascade also were observed. As the pressure increases, other Ar(4p) levels are formed. This information will be combined with our previous data on the growth and decay of the Ar(4p) emission to model both the radiative decay and collisional decay pathways of energy initially deposited in specific Ar(5p) levels.

After making the improvements in the laser which permitted wavelength resolved fluorescence to be observed, Bob Chang did experiments to monitor the appearance of the product Ar(4p') and Ar(4p) states with variation of pressure after pumping either the $2p_2$, $2p_3$ or $2p_4$ levels of the Ar(4p') manifold. Unfortunately, the $2p_1$ level cannot be produced because of the ΔJ restriction of dipole radiation. An example of the marked difference in the nature of the product states is displayed by comparing the product spectra when pumping the $2p_2$ and $2p_3$ levels as shown in Figure 2. If the $2p_2$ level is excited, the only emission observed even at 5.4 torr still originates from the $2p_2$ level. Since the quenching rate constant is large (Table 1), quenching must result in the <u>direct</u> conversion to the Ar (4s and 4s') levels. In contrast when pumping the $2p_3$ level, emission from the $2p_4$ and $2p_6$ levels, but not the $2p_5$ level, is easily observed even at 3.0 torr of pressure. Thus, for the $2p_3$ level quenching results in collisional cascade down the Ar(4p and 4p') manifold. The information regarding product level formation is summarized in Table 1.



ARGON $2p_2$
EXCITATION
5.4 TORR



ARGON $2p_3$
EXCITATION
3.0 TORR

Fig. 2

Table 1. Radiative lifetimes and quenching rate constants of Ar(4p') states.

Level Excited[a]	λ Laser(Å)	Radiative Lifetime(nsec)	Two-body Rate Constant[b]	Products and Branching Fractions	
$2p_2$ or $4p'[1/2]_1$	6965	26.5	5.34×10^{-11}	1s or (4s,4s') levels	100%
$2p_3$ or $4p'[3/2]_2$	7067	27.7	10.96×10^{-11}	$2p_4$ or $4p'[3/2]_1$	20%
				$2p_6$ or $4p[3/2]_2$	37%
				1s or (4s,4s') levels	43%
$2p_4$ or $4p'[3/2]_1$	7147	33.2	5.55×10^{-11}	1s or (4s,4s') levels	100%

a. The notations correspond to Paschen and Racah, respectively.

b. Units are $cm^3 atom^{-1} sec^{-1}$.

For the operating condition of the lasers, the most important levels, other than the Ar(4s and 4s'), probably will be the lower ones of the Ar(4p) manifold. However, we have been unable to produce these levels by laser pumping because of the long wavelength limitations of the dyes normally used with the N_2-pumped dye lasers. We recently have purchased some special red dyes and we will attempt to use these to pump some of the Ar(4p) set of levels. It is important to know whether or not these levels have large two-body rate constants for quenching to Ar(4s and 4s'). If the collisions mainly result in cascade to $2p_{10}$ and if that level is not readily quenched, significant concentrations of the $2p_{10}$ level may exist for discharge pumped lasers.

During the last quarter of the contract period, Dr. Chang will be completing the Ar(5p) work mentioned above, trying to use red sensitive dyes to produce the Ar(4p) levels and writing a manuscript describing these data. He also will be doing the experimental work on the Xe(6p') and Xe(7p) levels produced by laser pumping from the Xe($^3P_2$) level in the presence of argon buffer gas. We also should be able to observe the wavelength resolved fluorescence and to assign specific product states from the two-body collisions between Xe^{**} and Ar.

Vacuum U.V. Photolysis of XeF_2 and KrF_2 as Sources of XeF(B) and KrF(B).

In previous work (see attached reprint) we found that photolysis of XeF_2 provides an excellent source of XeF(B) and XeF(D). The addition of various quenching gases to the cell provides a means of studying the electronic quenching and a systematic study is in progress. The method also has been extended to KrF_2. The preliminary work shows that quenching studies are possible, but <u>much</u> more difficult to accomplish than for XeF_2. These difficulties arise for two reasons. First, KrF_2

is difficult to prepare in large quantities.[§] Second, the low bond energy of KrF_2 facilitates the reaction with F atoms, which, in fact, initiates a chain (non-explosive) decomposition.

$$F + KrF_2 \longrightarrow F_2 + KrF(X)$$

$$KrF(X) \longrightarrow Kr + F$$

The chain decomposition is particularly serious and requires that experiments be done with a flow of KrF_2 + diluent gas through the cell. The experiments, thus, require a large amount of KrF_2. The handling of KrF_2 also is more difficult than XeF_2 and complete passification of all lines is required. Since KrF_2 is even a stronger fluorinating agent than F_2, safety precautions must be exercised at all times.

The experiments are done in a vacuum U.V. photolysis cell. The output from a microwave powered resonance lamp enters the cell through a LiF window; the fluorescence is viewed perpendicularly to the photolysis axis with a monochromator. We have used Hg, Xe, CO, Kr and Ar resonance lamps. One should not use Xe or Kr resonance lamps with XeF_2 or KrF_2, respectively, for the following reason. There is likely to be Xe or Kr impurity remaining from the preparation, especially for KrF_2. Therefore, the resonance line is absorbed by the impurity atom, and one observes <u>sensitized</u> spectra rather than fluorescence spectra, e.g.

$$Kr + h\nu \longrightarrow Kr(^3P_1)$$

$$Kr(^3P_1) + KrF_2 \longrightarrow KrF(B) + Kr + F$$

$$KrF(B) \longrightarrow KrF(X) + h\nu$$

The KrF photodissociation spectra from the CO fourth-positive radiation is shown in Figure 3. Based upon $\Delta H_f^\circ(KrF_2)$, photodissociation with formation of KrF(B) requires 6eV of energy. The CO radiation does not provide a great deal of excess energy and the low vibrational

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[§]We employ low temperature photolysis of Kr + F_2 mixtures followed by low temperature distillation for purification.

distribution observed for KrF(B) is expected. The photolysis apparently produces very little of the KrF(C,3/2) state. The small amount of KrF(D,1/2) may result from some short wavelength impurity lines from the lamp. Providing that the flow of KrF_2 is stable, the KrF(B-X) fluorescence intensity seems to be comparable to that from photolysis of XeF_2. The results in Figure 3 demonstrate that KrF_2 can be photolyzed to give KrF(B). A flow system is mandatory to avoid the effects of the chain reaction and it is necessary to meter both KrF_2 and the quenching gas into the cell and to measure the partial pressures. Quenching experiments have been done with Xe because the very fast quenching permits low pressures of KrF_2 and Xe to be used. A flow of pure KrF_2 from a reservoir maintained at $\sim$ 15 torr was satisfactory. From the Stern-Volmer plot, the half-quenching pressure of KrF(B) by Xe is $\sim$ 2 torr. The product of the quenching has not been identified firmly. At the present time the cumulative evidence is that quenching of the KrF(B) state, with the vibrational distribution shown in Figure 3, <u>does not</u> proceed by the displacement reaction to give XeF(B) as a product.

After about six months of use, the extremely primitive gas handling vacuum system originally built to test the XeF_2 photolysis totally failed because of the corrosive nature of the XeF_2, KrF_2 and F_2 (used as a quenching gas). These vacuum system problems combined with the necessity of monitoring the partial pressures under flow operation for the KrF_2 experiments have forced Mr. Brashears to build a new and more satisfactory apparatus. As the last useful experiment with the old apparatus, Mr. Brashears did a survey of the emission spectra from sensitized reactions of $Xe(^3P_2)$ and $Kr(^3P_2)$ reacting with F_2, Cl_2 and other halogen containing molecules. Some of these spectra are shown

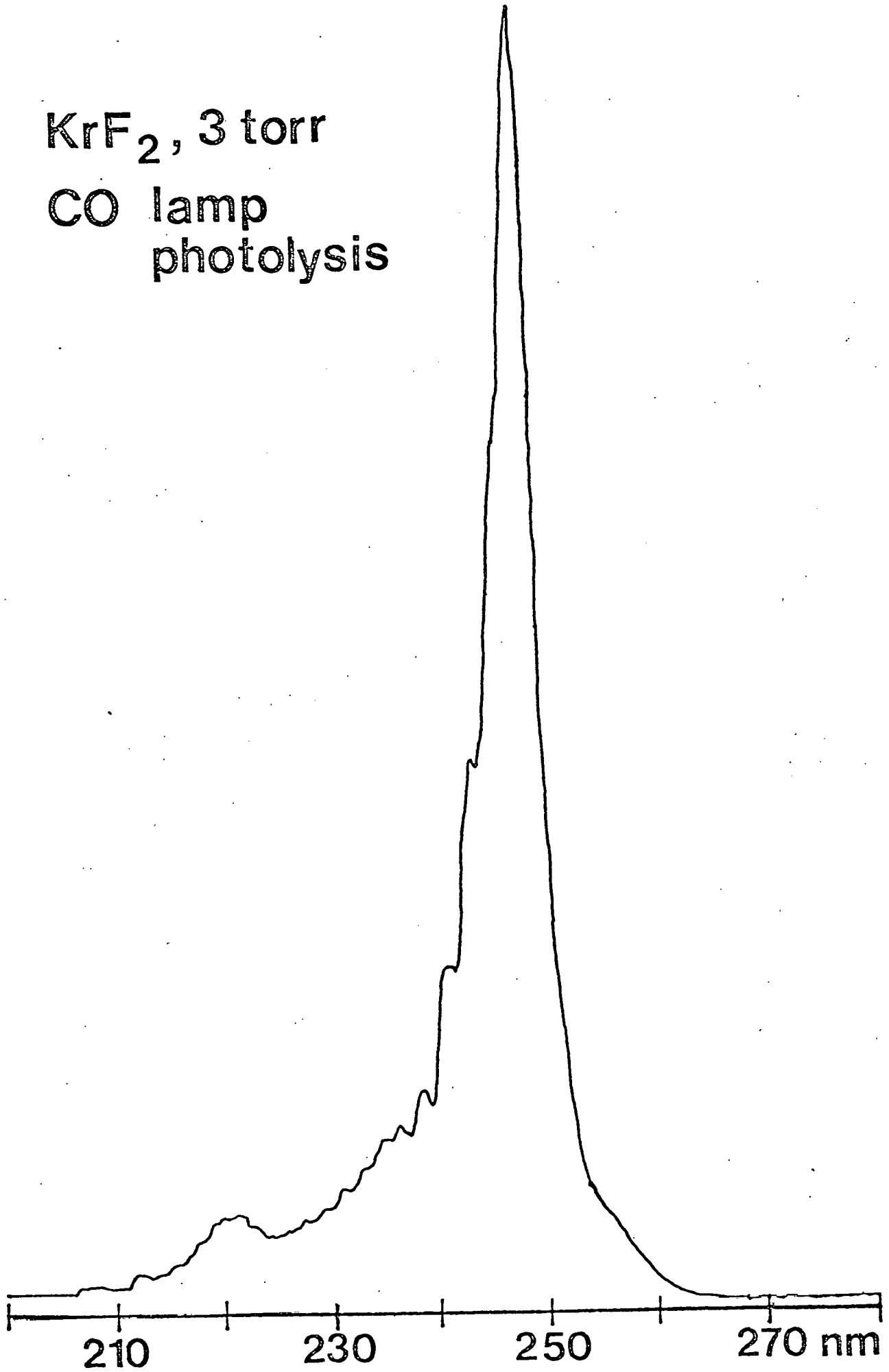


Fig. 3

in the renewal proposal. The rebuilding phase has taken place during June. The new vacuum monochromator also is being installed and will be used with the new apparatus to observe the output from the resonance lamps, which are notorious for strong impurity lines. With the new apparatus several atmospheres of pressure of quenching gas can be used to measure three-body quenching rates for KrF(B) and XeF(B) with rare gases that have slow two-body rates. With good fortune, the new apparatus should be ready for experiments by July 15. Even with this new apparatus, the work with KrF_2 will be difficult because of the difficulty of preparing and metering KrF_2. This new apparatus also will be used for the sensitization studies that are discussed in the proposal.

The reprint describing the photolysis of XeF_2 and the preliminary quenching measurements of XeF(B) complete this report.

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